

VBF-T1R1NOFORCE-DSH-01 - Technical Datasheet

VBF-T1R1NOFORCE-DSH-01 · Generation date: 2026-08-25 · Virtual design values (simulation)

Field	Value	Source
Rated capacity	5.0 Ah nominal (parameter set); 5.654 Ah verified (1C CC, 25 °C, to 2.5 V)	Chen2020 dump `Nominal cell capacity [A.h]`; `cell/r6`
Nominal voltage	2.5 and 4.2 V; discharge midpoint 4.1146 V	Chen2020 dump; `cell/r6_c7_energy.json:midpoint_v`
Rated energy	20.472 Wh (SV·I dt of 1C discharge)	`cell/r6_c7_energy.json:energy_wh`
Energy density	457.93 Wh/kg; 946.52 Wh/L (contract formula; electrolyte excluded from mass/volume)	`cell/r6_c7_energy.json`
Maximum continuous discharge	1C discharge (5.654 Ah); 5C not simulated	`cell/r6_c7_1c_spme.json`
Fast-charge capability	1C at 45 °C: 4.63 Ah (81.9 % SOC) in ~836 s (13.9 min); T_max 318.79 K; **no lithium plating** (anode potential min: 0.0032 V vs PMe4+0.005 V DFN)	Chen2020 dump; `cell/r6_c7_1c_spme.json`
DC resistance	4.029e-5 Ω (contract caliber, from 1C curve); power density 2.376 MW/kg (contract formula V_OC²/(4·DCR))	Chen2020 dump; `cell/r6_c7_energy.json:dc_r_ohm`
Operating temperature	Simulated conditions: 25 °C (1C, overcharge) and 45 °C (4C charge). Low temperature: Not simulated (-20 °C not in scope available)	Chen2020 dump; `cell/r6_c7_energy.json`
Cycle life	Not simulated (aging model available on Chen2020 but not in task scope; must not fabricate)	honest scope statement
Safety determination	Overcharge to 4.7 V: thermal runaway **not triggered** (three-side-reaction ODE, T_max 298.22 K, dT/dt max 18.6 °C/s at 4C temperature rise 0.64 s above 45 °C)	Chen2020 dump; `cell/r6_c7_energy.json`
Dimensions and mass	Electrode 65 mm x 1580 mm (unwound); stack thickness 210.6 μm; shell dimensions Not provided. Mass 4.765200 dump; electrolyte excluded; 51.448 g incl.	Chen2020 dump; `cell/r6_c7_energy.json`
System	NMC811 / graphite, single-crystal cathode 1.2 μm, fine graphite anode 0.8 μm, EC/EMC + LiPF6 (sigma 1.8e-5 Pa·s, perm 0.5)	Chen2020 dump; `cell/r6_c7_energy.json`